Absolutely. Below are **80/20 interview-focused notes** for this chapter. I'll separate **must-know concepts**, **implementation importance**, **intuition**, **visual diagrams**, and **interview questions** so you can focus on the smallest set of mathematics that gives you the most ML/interview value.

Chapter 2 — Geometric Structure of Vector Spaces

80/20 Interview Notes for Machine Learning

The big picture

The entire chapter can be compressed into this chain:

Vector Space

|

▼

Inner Product (dot product)

|

├──────────► Similarity / Angles / Orthogonality

|

▼

Norm

|

▼

Distance

|

▼

Geometry of data

|

├── Clustering

├── Similarity

├── Regularization

├── Projections

├── PCA / dimensionality reduction

└── ML optimization

⭐ **The most important 80/20 ideas**

If you have limited time, prioritize these:

Priority	Concept	Interview Importance	Implementation
🔴 P0	Norm / magnitude	Very High	Yes
🔴 P0	Euclidean distance	Very High	Yes
🔴 P0	Manhattan distance	High	Yes
🔴 P0	Inner/dot product	**Extremely High**	Yes
🔴 P0	Cosine similarity	**Extremely High**	Yes
🔴 P0	Orthogonality	Very High	Yes
🔴 P0	Projection	**Extremely High**	Yes
🔴 P0	Orthonormal basis	High	Understand
🔴 P0	Gram-Schmidt intuition	High	Understand + basic implementation
🔴 P0	L1/L2 regularization	**Extremely High for ML**	Yes
🟠 P1	Cauchy-Schwarz	High mathematically	Understand
🟠 P1	Triangle inequality	High	Understand
🟠 P1	Orthogonal complement	Medium-High	Understand
🟠 P1	Parseval identity	Medium	Understand
🟡 P2	Function-space supremum norm	Low	No
🟡 P2	Formal proofs	Medium for math interviews	Usually no
🟡 P2	Detailed limit proof of ∞-norm	Low	No

---

1. Vector as a Point / Feature Representation

In ML, a data sample is commonly represented as a vector.

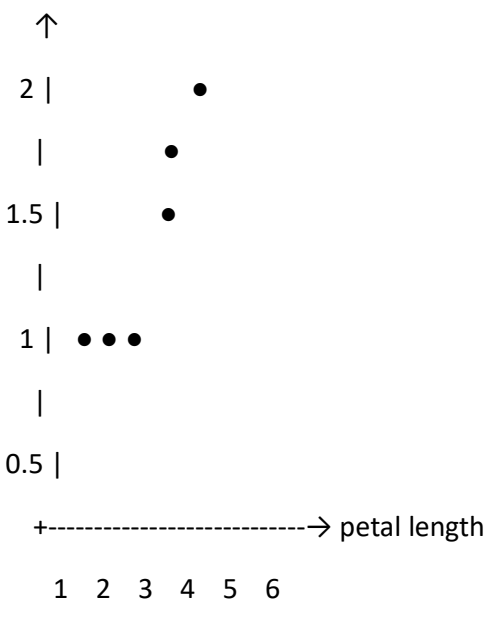
For example:

Iris flower

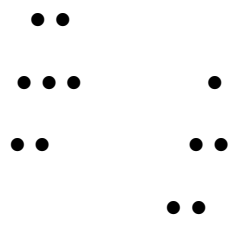
x = [petal_length, petal_width]

x = [5.1, 1.8]

Geometrically:



If points naturally form groups:



we can suspect different classes/clusters.

Interview takeaway

Distance provides a way to quantify how close or different data points are.

This is fundamental to:

- clustering
- nearest neighbors
- anomaly detection
- recommendation
- embeddings
- similarity search

---

2. Norm — Magnitude of a Vector ⭐ P0

Definition

A **norm** measures the magnitude/length of a vector.

A function

$$\|\cdot\|: V \to [0, \infty)$$

is a norm if:

1. Positive definiteness

$$\| x \| \geq 0$$

and

$$\| x \| = 0 \iff x = 0$$

2. Positive homogeneity

$$\| cx \| = | c | \| x \|$$

3. Triangle inequality

$$\| x + y \| \leq \| x \| + \| y \|$$

Intuition

Think of:

x

●

/

/

/

/

●

0

The norm is the length of the arrow:

||x|| = length of x

---

3. L2 / Euclidean Norm ⭐⭐⭐

The most important norm for ML.

For

$$x = (x_1, x_2, \dots, x_n)$$

the Euclidean norm is:

$$\boxed{\| x \|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}}$$

Example:

$$x = (3, 4)$$

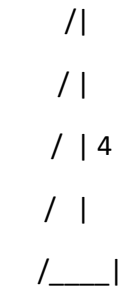
Then:

$$\| x \|_2 = \sqrt{3^2 + 4^2} = 5$$

Visual:

(3,4)

●



length = 5

This is simply the **Pythagorean theorem**.

Why ML cares

L2 appears everywhere:

- Euclidean distance
- MSE
- L2 regularization
- least squares
- neural-network weight penalties
- optimization

---

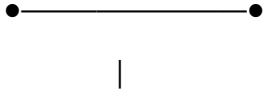
4. L1 / Manhattan Norm ⭐ ⭐

$$\boxed{\| x \|_1 = \sum_i | x_i |}$$

For:

$$x = (3, -4)$$
$$\| x \|_1 = | 3 | + | -4 | = 7$$

Think of navigating a city grid:



|

|

•

You cannot travel diagonally.

So:

$$d = 3 + 4 = 7$$

rather than:

$$\sqrt{3^2 + 4^2} = 5$$

---

5. Lp Norm ⭐

General form:

$$\boxed{\| x \|_p = \left(\sum_{i=1}^{n} | x_i |^p \right)^{1/p}}$$

Important special cases:

$$p = 1 \Rightarrow L1$$
$$p = 2 \Rightarrow L2$$
$$p \to \infty \Rightarrow L_\infty$$

---

6. L∞ Norm

$$\boxed{\| x \|_\infty = \max_i | x_i |}$$

Example:

$$x = (3, -, 72)$$

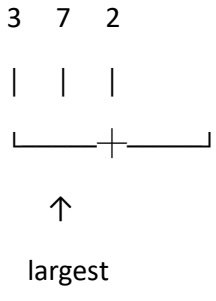
Therefore:

$$\| x \|_\infty = 7$$

Intuition

It only cares about the **largest absolute component**.

x = [3, -7, 2]



Interview importance

Know the definition and relationship with L_p.

You generally don't need to memorize the detailed limit proof unless you're specifically preparing for mathematical interviews.

---

7. Distance ⭐ ⭐ ⭐

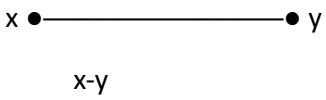
A very important relationship:

$$\boxed{d(x, y) = \| x - y \|}$$

In other words:

Distance = norm of the difference vector.

Visual:



distance = ||x-y||

Euclidean distance

$$\boxed{d(x,y) = \sqrt{\sum_i (x_i - y_i)^2}}$$

Example:

$$x = (1, 2)$$
$$y = (4, 6)$$

Then:

$$x - y = (-, 3 - 4)$$

and:

$$d(x,y) = \sqrt{9 + 16} = 5$$

Manhattan distance

$$\boxed{d(x,y) = \sum_i | x_i - y_i |}$$

For the same example:

$$d(x,y) = 3 + 4 = 7$$

---

8. Norm vs Distance — Very Important Interview Distinction

This is a common interview question.

Norm

Measures magnitude from the origin:

$$\boxed{\| x \|}$$

x

•

/

/

/

●

0

Distance

Measures separation between two vectors:

$$\boxed{d(x,y) = \| x - y \|}$$

x ●——————● y

distance

One-line answer

A norm measures the magnitude of a vector, while a metric measures the distance between two points; a norm induces a metric through $d(x,y) = \| x - y \|$.

---

9. Metric Properties ⭐ ⭐

A metric must satisfy:

1. Non-negativity / positive definiteness

$$d(x,y) \geq 0$$

and

$$d(x,y) = 0 \iff x = y$$

2. Symmetry

$$d(x,y) = d(y,x)$$

3. Triangle inequality

$$d(x,z) \leq d(x,y) + d(y,z)$$

Visual:

y

●



direct path x → z

cannot be longer than

x → y → z

⚠️ **Important correction**

The supplied text contains a typo near the metric definition:

It says that positive definiteness gives $d(x, y) = 1 \Rightarrow x = y$.

That should be:

$$\boxed{d(x, y) = 0 \Rightarrow x = y}$$

---

10. Inner Product / Dot Product ⭐ ⭐ ⭐ ⭐ ⭐

This is one of the **highest-priority concepts in the entire chapter**.

For:

$$x = (x_1, \ldots, x_n)$$

and

$$y = (y_1, \ldots, y_n)$$

the dot product is:

$$\boxed{\langle x, y \rangle = \sum_i x_i \, y_i}$$

For example:

$$x = (1, 2, 3)$$

$$y = (4, 5, 6)$$

then:

$$x \cdot y = 1(4) + 2(5) + 3(6)$$
$$= 4 + 10 + 18 = 32$$

---

11. Why Dot Product Matters So Much

The dot product gives us information about:

Dot Product

|

├— similarity

├— angle

├— orthogonality

├— projection

├— cosine similarity

├— vector magnitude

└— neural-network computations

It is also fundamental to:

- linear regression
- neural networks
- embeddings
- attention
- PCA
- SVMs
- similarity search

---

12. Dot Product and Angle ⭐ ⭐ ⭐ ⭐ ⭐

The key equation:

$$\boxed{x \cdot y = \| x \| \| y \| \cos\theta}$$

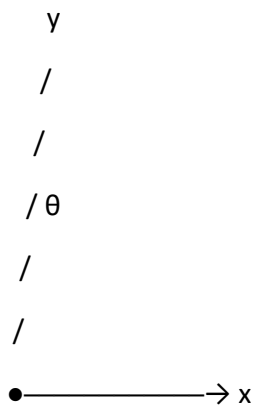
Therefore:

$$\boxed{\cos\theta = \frac{x \cdot y}{\| x \| \| y \|}}$$

and:

$$\boxed{\theta = \cos^{-1}\left(\frac{x \cdot y}{\| x \| \| y \|}\right)}$$

Visual



The dot product tells us how strongly two vectors point in the same direction.

---

13. Sign of Dot Product ⭐ ⭐ ⭐ ⭐ ⭐

This is an excellent interview intuition question.

Same/general direction

$$x \cdot y > 0$$

→

↗

Angle:

$$0^\circ < \theta < 90^\circ$$

Orthogonal

$$x \cdot y = 0$$

↑

|

|

└——→

Angle:

$$\theta = 90^\circ$$

Opposite direction

$$x \cdot y < 0$$

← →

Angle:

$$90^\circ < \theta \leq 180^\circ$$

---

14. Orthogonality ⭐ ⭐ ⭐ ⭐ ⭐

Two vectors are orthogonal if:

$$\boxed{\langle x, y \rangle = 0}$$

For example:

$$x = (1, 0)$$
$$y = (0, 1)$$

Then:

$$x \cdot y = 0$$

Therefore:

$$x \perp y$$

ML intuition

Orthogonality means:

The vectors have zero dot-product interaction under the chosen inner product.

Don't oversimplify this into "they are always independent." Orthogonality implies linear independence for nonzero orthogonal vectors, but "orthogonal = no information whatsoever" is an intuition, not the complete mathematical statement.

---

15. Inner Product Generates the L2 Norm ⭐ ⭐ ⭐ ⭐ ⭐

A very important relationship:

$$\boxed{\| x \| = \sqrt{\langle x, x \rangle}}$$

For ordinary Euclidean vectors:

$$\langle x, x \rangle = x_1^2 + x_2^2 + \cdots + x_n^2$$

Therefore:

$$\| x \|_2 = \sqrt{x_1^2 + \cdots + x_n^2}$$

The hierarchy

Inner Product

|

▼

<x,x>

|

▼

sqrt(<x,x>)

|

▼

Norm

And:

Norm

|

▼

||x-y||

|

▼

Distance

This is one of the central ideas of the chapter.

---

16. Cauchy-Schwarz Inequality ⭐⭐⭐

The theorem:

$$\boxed{|\langle x, y\rangle| \leq \|x\| \|y\|}$$

Equivalently:

$$\boxed{|\langle x, y\rangle|^2 \leq \langle x, x\rangle \langle y, y\rangle}$$

Why it matters

It guarantees:

$$-1 \leq \frac{x \cdot y}{\|x\| \|y\|} \leq 1$$

which makes cosine similarity valid.

Interview level

Know:

- statement
- intuition
- relationship to cosine similarity

You usually don't need to reproduce the quadratic-discriminant proof unless specifically asked.

---

17. Projection ⭐ ⭐ ⭐ ⭐ ⭐

Projection is **very important for ML**.

Suppose we want to find how much of $x$points in the direction of y.

The projection of $x$onto y:

$$\boxed{\text{proj}_y(x) = \frac{\langle x, y \rangle}{\langle y, y \rangle} y}$$

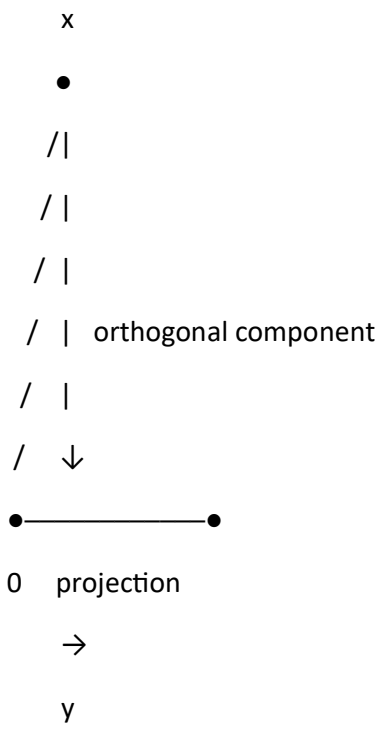
Since:

$$\langle y, y \rangle = \| y \|^2$$

we can also write:

$$\boxed{\text{proj}_y(x) = \frac{x \cdot y}{\| y \|^2} y}$$

Visual



We decompose:

$$\boxed{x = x_p + x_o}$$

where:

- x_p= component parallel to y
- x_o= component orthogonal to y

and:

$$x_o \perp y$$

---

18. Why Projection Matters in ML

Projection appears in many important concepts:

- linear regression
- least squares
- PCA
- dimensionality reduction
- feature spaces
- optimization
- signal processing
- orthogonal decomposition

Key intuition

Projection extracts the component of one vector that lies in another vector's direction/subspace.

---

19. Cosine Similarity ⭐⭐⭐⭐⭐

One of the most practically important formulas from the chapter.

$$\boxed{\cos(x, y) = \frac{x \cdot y}{\| x \| \| y \|}}$$

It measures **directional similarity**, not absolute distance.

Example

Suppose:

$$x = (1, 1)$$
$$y = (2, 2)$$

Then:

$$x \cdot y = 4$$
$$\| x \| = \sqrt{2}$$
$$\| y \| = 2\sqrt{2}$$

Therefore:

$$\cos(x, y) = 1$$

They point in exactly the same direction.

But:

$$d(x, y) = \sqrt{2}$$

So they are not the same point.

---

20. Distance vs Cosine Similarity ⭐⭐⭐⭐⭐

This is an excellent interview distinction.

Distance

Answers:

How far apart are these vectors?

x ●——————————● y

Cosine similarity

Answers:

How similarly are these vectors oriented?

x ↗

/

/

/ y ↗

Two vectors can have:

HIGH cosine similarity

+

LARGE Euclidean distance

Example:

$$x = (1, 1)$$
$$y = (100, 100)$$

They point in exactly the same direction, so cosine similarity is:

$$1$$

but they are far apart.

---

21. Unit Vectors

A unit vector has norm:

$$\boxed{\| x \| = 1}$$

We can normalize a nonzero vector:

$$\boxed{\hat{x} = \frac{x}{\| x \|}}$$

Visual:

Original vector

•

/

/

/

•

0

↓ normalize

•

/

/

•

0

length = 1

This is extremely important in ML because cosine similarity is essentially the dot product of normalized vectors.

---

22. Orthonormal Basis ⭐ ⭐ ⭐ ⭐

A basis is **orthonormal** if:

Different basis vectors are orthogonal

$$\langle v_i, v_j \rangle = 0 \; i \neq j$$

Each has unit length

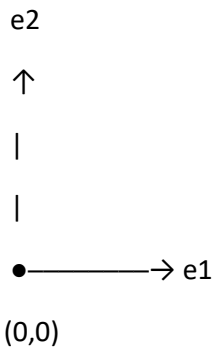
$$\| v_i \| = 1$$

Combined:

$$\boxed{\langle v_i, v_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}}$$

Example in $\mathbb{R}^2$:

$$e_1 = (1, 0)$$
$$e_2 = (0, 1)$$



---

23. Why Orthonormal Bases Are Useful

Suppose:

$$x = \sum_i c_i \, v_i$$

With an arbitrary basis, finding $c_i$can require solving a system of equations.

With an orthonormal basis:

$$\boxed{c_i = \langle x, v_i \rangle}$$

Therefore:

$$\boxed{x = \sum_i \langle x, v_i \rangle v_i}$$

This is extremely elegant.

Intuition

Each coefficient tells us:

"How much of $x$points in the direction of v_i?"

---

24. Parseval's Identity ⭐ ⭐ ⭐

For an orthonormal basis:

$$x = \sum_i c_i \, v_i$$

then:

$$\boxed{\| \, x \, \|^2 = \sum_i c_i^2}$$

This says:

The squared magnitude of a vector equals the sum of squared coordinates in an orthonormal coordinate system.

For the standard basis, this becomes the familiar:

$$x_1^2 + x_2^2 + \cdots + x_n^2$$

Interview priority

Understand the intuition.

Don't spend large amounts of study time memorizing the proof.

---

25. Gram-Schmidt Orthogonalization ⭐ ⭐ ⭐ ⭐

Problem

You have arbitrary basis vectors:

v1

↗

v2

↗

but you want:

e1 ↑

|

|

└——→ e2

an orthogonal/orthonormal basis.

Core idea

Take each vector and **remove the components already explained by previous vectors**.

Step 1

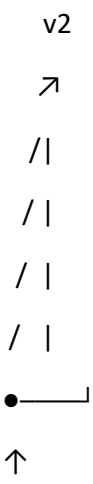
$$e_1 = v_1$$

Step 2

Remove the projection of $v_2$onto e_1:

$$\boxed{e_2 = v_2 - \text{proj}_{e_1}(v_2)}$$

Visual:



remaining part



Step 3

Remove the components of v_3 along e_1 and e_2:

$$\boxed{e_3 = v_3 - \text{proj}_{e_1}(v_3) - \text{proj}_{e_2}(v_3)}$$

Continue.

---

26. Gram-Schmidt — Interview Intuition

Memorize this sentence:

Gram-Schmidt converts linearly independent vectors into an orthonormal basis by repeatedly subtracting the projections onto previously generated basis vectors and then normalizing.

That is much more useful than memorizing the entire proof.

---

27. Gram-Schmidt Algorithm

For vectors:

$$v_1, v_2, \dots, v_n$$

Orthogonalization

$$u_1 = v_1$$
$$u_2 = v_2 - \text{proj}_{u_1}(v_2)$$
$$u_3 = v_3 - \text{proj}_{u_1}(v_3) - \text{proj}_{u_2}(v_3)$$

and so on.

Normalize

$$\boxed{e_i = \frac{u_i}{\| u_i \|}}$$

Result:

$$e_1, e_2, \dots, e_n$$

are orthonormal.

---

28. Linearly Dependent Input

Suppose:

$$v_2 = cv_1$$

Then $v_2$contains no new direction.

Projection removes everything:

$$v_2 - \text{proj}_{v_1}(v_2) = 0$$

So Gram-Schmidt produces:

v1 → useful direction

v2 → duplicate direction

↓

e1 → valid

e2 → 0

Important intuition

Gram-Schmidt exposes redundant directions.

---

29. Orthogonal Complement ⭐ ⭐ ⭐

For a subspace S:

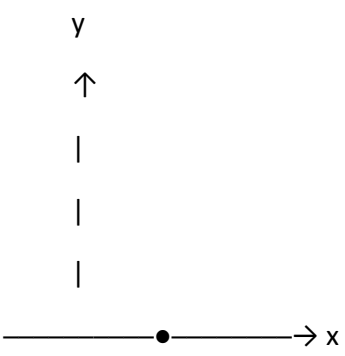
$$\boxed{S^{\perp} = \{x : x \perp s \text{ for every } s \in S\}}$$

In words:

$S^{\perp}$contains every vector perpendicular to every vector in S.

Example

In $\mathbb{R}^2$:



If:

$$S = \text{x-axis}$$

then:

$$S^{\perp} = \text{y-axis}$$

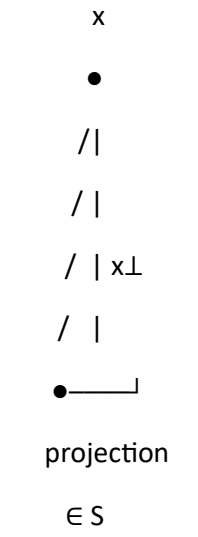
---

30. Decomposition Using Orthogonal Complement

For a finite-dimensional inner product space:

$$\boxed{V = S + S^{\perp}}$$

Intuitively:



Every vector can be decomposed into:

$$\boxed{x = x_S + x_{S^\perp}}$$

where:

$$x_S \in S$$

and:

$$x_{S^\perp} \in S^\perp$$

This is a very important idea behind projections and least-squares methods.

---

31. Regularization — Extremely Important for ML ⭐⭐⭐⭐⭐

The chapter connects norms to controlling model complexity.

Suppose:

$$f(x) = q_0 + q_1 x + q_2 x^2 + \cdots + q_m x^m$$

Instead of minimizing only:

$$MSE$$

we minimize:

$$\boxed{Loss = MSE + \lambda \parallel q \parallel_p}$$

where:

- q= model parameters
- λ= regularization strength
- $p = 1$→ L1 regularization
- $p = 2$→ L2 regularization

---

32. L1 vs L2 Regularization ⭐⭐⭐⭐⭐

L1

$$\boxed{Loss = MSE + \lambda \parallel w \parallel_1}$$

Encourages many weights to become exactly zero.

Therefore:

Before:

[w1, w2, w3, w4, w5]

After L1:

[w1, 0, 0, w4, 0]

Main intuition

Feature/parameter sparsity.

L2

$$\boxed{Loss = MSE + \lambda \parallel w \parallel_2^2}$$

Encourages weights to become smaller rather than typically forcing them exactly to zero.

Before:

[10, 8, 6, 4]

After:

[3.2, 2.5, 1.8, 1.1]

Main intuition

Weight shrinkage.

---

33. ⚠️ Important L2 Detail for Interviews

The chapter writes a general regularization term:

$$\lambda \parallel q \parallel_p$$

In practical ML, you will very often encounter:

$$\boxed{\lambda \parallel w \parallel_2^2}$$

for L2 regularization rather than simply $\lambda \parallel w \parallel_2$.

Similarly, L1 is often written:

$$\lambda \parallel w \parallel_1$$

Know both notation conventions.

---

34. Implementation — What You Actually Need to Code ⭐⭐⭐⭐⭐

For your ML engineering path, these concepts should **not remain purely theoretical**.

You should be able to implement the following from scratch using Python/NumPy.

Implementation 1 — L2 norm

```
import numpy as np


x = np.array([3, 4])


norm = np.linalg.norm(x)


print(norm)

# 5.0
```

From scratch

```
norm = np.sqrt(np.sum(x ** 2))
```

---

35. Implementation 2 — L1 norm

```
x = np.array([3, -4])


l1 = np.sum(np.abs(x))


print(l1)

# 7
```

---

36. Implementation 3 — Euclidean Distance

```
x = np.array([1, 2])

y = np.array([4, 6])


distance = np.linalg.norm(x - y)


print(distance)

# 5.0
```

From scratch:

```
distance = np.sqrt(np.sum((x - y) ** 2))
```

---

37. Implementation 4 — Dot Product ⭐ ⭐ ⭐ ⭐ ⭐

```
x = np.array([1, 2, 3])

y = np.array([4, 5, 6])


dot = np.dot(x, y)


print(dot)

# 32
```

From scratch:

dot = np.sum(x * y)

You should absolutely understand this implementation.

---

38. Implementation 5 — Cosine Similarity ⭐ ⭐ ⭐ ⭐ ⭐

```
def cosine_similarity(x, y):
    return np.dot(x, y) / (
        np.linalg.norm(x) * np.linalg.norm(y)
    )
```

Example:

```
x = np.array([1, 1])
y = np.array([2, 2])


print(cosine_similarity(x, y))
# 1.0
```

Production consideration

If either vector is zero:

$$\| x \| = 0$$

the formula divides by zero.

So production code should handle zero vectors.

---

39. Implementation 6 — Projection ⭐ ⭐ ⭐ ⭐ ⭐

```
def projection(x, y):
    return (np.dot(x, y) / np.dot(y, y)) * y
```

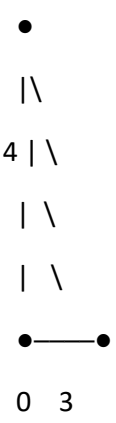
Example:

```
x = np.array([3, 4])
y = np.array([1, 0])
```

```
print(projection(x, y))
# [3, 0]
```

Visual:

x = (3,4)



projection = (3,0)

---

40. Implementation 7 — Gram-Schmidt

A basic implementation is worth understanding:

```
import numpy as np

def gram_schmidt(vectors):
    orthogonal = []

    for v in vectors:
        u = v.astype(float).copy()

        for prev in orthogonal:
            u -= (
```

```
            np.dot(v, prev)
            / np.dot(prev, prev)
        ) * prev

    orthogonal.append(u)

orthonormal = [
    u / np.linalg.norm(u)
    for u in orthogonal
]

return orthonormal
```

Important production note

For numerical linear algebra, classical Gram-Schmidt can suffer from numerical instability. In production numerical work, **QR decomposition / modified Gram-Schmidt** is often preferable.

For interviews, however, you should first master the conceptual algorithm.

---

41. One Very Important Connection: Dot Product → Norm → Distance

Memorize this structure:

$$\boxed{\langle x, y \rangle}$$

gives the inner product.

Then:

$$\boxed{\| x \| = \sqrt{\langle x, x \rangle}}$$

gives the L2 norm.

Then:

$$\boxed{d(x,y) = \| x - y \|}$$

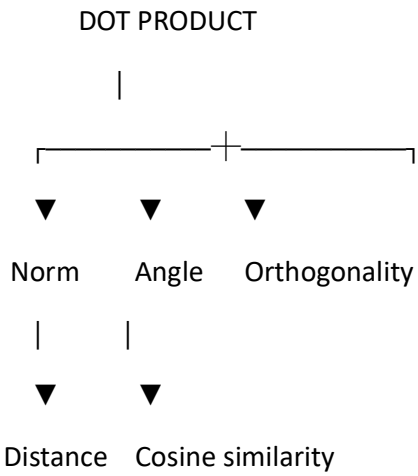
gives Euclidean distance.

And:

$$\boxed{\cos(x,y) = \frac{\langle x,y \rangle}{\| x \| \| y \|}}$$

gives cosine similarity.

So:



This is probably the **single most important conceptual map** in this chapter.

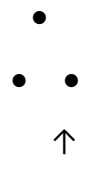
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42. Machine Learning Applications

1. K-Nearest Neighbors

Find points with small distance:

$$d(x, x_i)$$



Nearest points influence the prediction.

---

2. Clustering

Algorithms such as K-Means depend heavily on distances.

For example:

$$\| x - \mu_k \|_2^2$$

measures distance from point $x$to cluster center μ_k.

---

3. Embeddings

Suppose text/images/products are represented as vectors:

"machine learning"

↓

[0.21, -0.17, 0.82, ...]

Similarity can be measured using:

cosine similarity

This is fundamental to:

- semantic search
- recommendation
- RAG
- vector databases
- retrieval systems

---

4. Neural Networks

A basic neuron computes:

$$z = w^T x + b$$

and:

$$w^T x = \sum_i w_i x_i$$

That $w^T x$is simply a **dot product**.

So the dot product isn't merely abstract mathematics.

It is directly inside neural-network computation.

---

43. A Deeper ML Connection

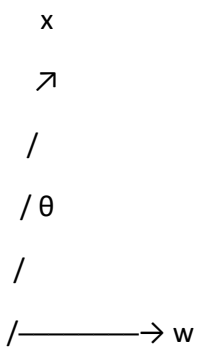
Think of:

$$w^T x$$

as asking:

"How strongly does input $x$align with the learned direction w?"

Visual:



Then:

$$w^T x = \| w \| \| x \| \cos\theta$$

So a neuron is closely connected to the geometry of vector spaces.

---

44. What You Should NOT Overstudy — 80/20

You don't need equal depth everywhere.

🔴 **Master**

These deserve real understanding + coding:

1. Norm
2. L1/L2/L∞
3. Euclidean distance
4. Manhattan distance
5. Dot product
6. Angle
7. Orthogonality
8. Cosine similarity
9. Projection
10. Normalization
11. Orthonormal basis
12. Gram-Schmidt
13. L1/L2 regularization

🟠 **Understand**

Know intuition + formulas:

14. Cauchy-Schwarz
15. Triangle inequality
16. Orthogonal complement
17. Parseval's identity
18. Norm-induced metric

🟡 **Read once / low priority**

19. Function-space supremum norm
20. Detailed proof of the $p \to \infty$limit
21. Detailed formal proofs of every theorem
22. The function-space technicality concerning equality almost everywhere

These are mathematically interesting but have much lower ROI for a practical ML/backend engineering interview.

---

45. Interview Cheat Sheet

Norm

Q: What is a norm?

A norm measures the magnitude of a vector and satisfies positive definiteness, positive homogeneity, and the triangle inequality.

L2 norm

$$\boxed{\| x \|_2 = \sqrt{\sum_i x_i^2}}$$

L1 norm

$$\boxed{\| x \|_1 = \sum_i | x_i |}$$

L∞ norm

$$\boxed{\| x \|_\infty = \max_i | x_i |}$$

Distance

$$\boxed{d(x, y) = \| x - y \|}$$

Dot product

$$\boxed{x \cdot y = \sum_i x_i y_i}$$

Angle

$$\boxed{\cos\theta = \frac{x \cdot y}{\| x \| \| y \|}}$$

Orthogonality

$$\boxed{x \perp y \iff x \cdot y = 0}$$

Projection

$$\boxed{\text{proj}_y(x) = \frac{x \cdot y}{y \cdot y} y}$$

Cosine similarity

$$\boxed{\text{cosine}(x, y) = \frac{x \cdot y}{\| x \| \| y \|}}$$

Orthonormal basis

$$\boxed{v_i \cdot v_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}}$$

Coefficient in an orthonormal basis

$$\boxed{c_i = x \cdot v_i}$$

Gram-Schmidt

$$\boxed{\text{new vector} = \text{original vector} - \text{previous projections}}$$

then normalize.

---

46. Most Likely Interview Questions

Basic

1. What is a norm?

2. What is the difference between norm and distance?
3. Explain L1 vs L2 norm.
4. What is Euclidean distance?
5. What is Manhattan distance?
6. What is a dot product?
7. What does a positive/zero/negative dot product mean?
8. What does orthogonal mean?
9. What is cosine similarity?
10. Difference between cosine similarity and Euclidean distance?

Intermediate

11. Why is cosine similarity useful for embeddings?
12. Why can two vectors have cosine similarity 1 but nonzero distance?
13. How do you normalize a vector?
14. Derive Euclidean norm from the dot product.
15. Explain projection geometrically.
16. What is an orthonormal basis?
17. Why are orthonormal bases useful?
18. Explain Gram-Schmidt.
19. What happens if Gram-Schmidt receives linearly dependent vectors?
20. What is the orthogonal complement?

ML-focused

21. Why does L2 regularization help prevent overfitting?
22. Why does L1 regularization encourage sparsity?
23. Where does the dot product appear in neural networks?
24. Why is vector similarity important in RAG?
25. When would you prefer cosine similarity over Euclidean distance?
26. What happens when a vector has zero magnitude during cosine similarity?
27. Why does feature scaling matter when using distance-based algorithms?

28. How does distance affect K-Means?

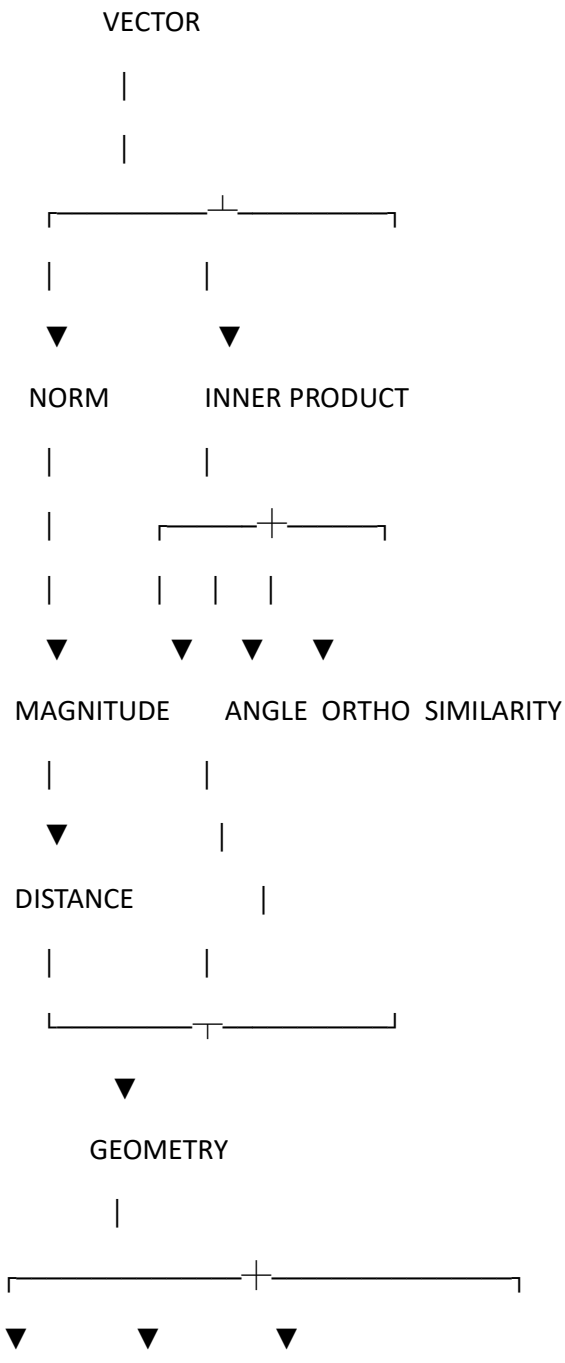
29. How is projection related to least squares?

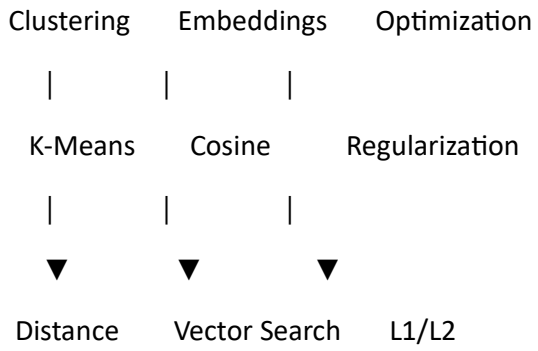
30. Where does orthogonality appear in PCA?

---

47. One Visual Mental Model for the Whole Chapter

Keep this in your head:





---

48. Final 80/20 Summary

If you remember only **10 things**, remember these:

1.

$$\boxed{\| x \| = \text{magnitude}}$$

2.

$$\boxed{d(x,y) = \| x - y \|}$$

3.

$$\boxed{x \cdot y = \sum_i x_i y_i}$$

4.

$$\boxed{x \cdot y = \| x \| \| y \| \cos\theta}$$

5.

$$\boxed{x \perp y \iff x \cdot y = 0}$$

6.

$$\boxed{\cos(x,y) = \frac{x \cdot y}{\| x \| \| y \|}}$$

7.

$$\boxed{\text{proj}_y(x) = \frac{x \cdot y}{y \cdot y} y}$$

8.

$$\boxed{\| x \| = \sqrt{\langle x, x \rangle}}$$

9.

$$\boxed{\text{Orthonormal basis} = \text{orthogonal} + \text{unit length}}$$

10.

$$\boxed{\text{Gram-Schmidt} = \text{remove previous projections} + \text{normalize}}$$

And the **ML connection** to remember is:

Vectors

↓

Geometry

↓

Distance + Similarity + Projection

↓

ML

- ├— KNN
- ├— K-Means
- ├— Embeddings
- ├— RAG / Vector Search
- ├— Neural Networks
- ├— Least Squares
- ├— PCA
- └— L1/L2 Regularization

🎯 **Interview priority**

For your **Applied Mathematics of ML → AI/ML Engineering** preparation, I would spend approximately:

60% of your chapter time: dot product, norms, distances, cosine similarity, projections, orthogonality, regularization
25%: orthonormal bases + Gram-Schmidt + orthogonal complement
10%: Cauchy-Schwarz + Parseval + metric properties
5%: function-space examples and detailed proofs

The goal isn't to memorize the chapter. The real mastery test is whether you can **look at a vector problem and immediately decide whether magnitude, distance, angle, similarity, projection, or orthogonality is the right mathematical tool.**